

Sepehr Saeedpour

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Education

- **École Normale Supérieure (ENS) – PSL University** Paris, France
Normalien and M.Sc. in Cognitive Science (Modeling, Neurotheory and AI Track) 2024 – present
Thesis: Inferring Head-Direction Dynamics using Gaussian Process State Space Models — **Highest grade in cohort**
- **University of Tehran – School of Electrical and Computer Engineering** Tehran, Iran
B.Sc. in Electrical Engineering (Control Systems Track) 2018 – 2023
Thesis: Interindividual Differences in Learning to Act or Avoid — **Best Undergraduate Thesis Award**

Research Experience

- **Sorbonne Université — Institut des Systèmes Intelligents et de Robotique (ISIR)** Paris, France
Research Assistant Aug 2025 – present
Supervisor: Dr. Heike Stein
 - Modeling population-level neural dynamics in the head-direction system using Gaussian-process state-space models (GPSSMs), treating observed neural activity as noisy evidence about a low-dimensional latent angular state evolving under nonlinear dynamics; model fit to *mouse* and *Drosophila* data. (pre-print in preparation.)
- **École Normale Supérieure (ENS) — Human Reinforcement Learning Lab** Paris, France
Research Intern Feb 2024 – June 2025
Co-supervisors: Prof. Stefano Palminteri, Dr. Hernan Anllo
 - Investigated the computational mechanisms of experiential knowledge transmission (e.g., teaching) by designing and implementing an online behavioural experiment ([demo](#)), recruiting and testing 50 participants.
- **Ludwig Maximilians Universität München — Crowd Cognition Group** Munich, Germany & Remote
Undergraduate Research Assistant June 2021 – August 2024
Co-supervisors: Dr. Bahador Bahrami, Prof. Dr. Ophelia Deroy
 - Fitted reinforcement-learning models to behavioral data from a Go/NoGo task to infer latent cognitive parameters underlying Pavlovian-instrumental interactions; published results as first author.
 - Quantified goal-directed vs. habitual control in social decision-making by fitting model-based and model-free reinforcement-learning models to behavioral data and co-authoring the manuscript.
 - Developed real-time eye-tracking pipelines for interactive environments and characterized gaze dynamics using time-series analysis and visual saliency representations.

Publications

- Zhong, Z., Prystawski, B., Wu, S. A., Fascendini, B., **Saeedpour, S.**, Austerweil, J. L. (2026). Interpretational alignment: How agents learn from physical guidance depends on how they interpret it. *Accepted in CogSci Proceedings 2026*.
- Aslani, A., **Saeedpour, S.**, Ferreiro, D., Bahrami, B., Deroy, O. (2026). Do touch screens work? Assessing individual engagement with video, touch, and phone screens in museums using mobile eye-tracking. *Under review, Visitor Studies*.
- **Saeedpour, S.**, Miandari, M., Deroy, O., Bahrami, B. (2023). Interindividual differences in Pavlovian influence on learning are consistent. *Royal Society Open Science*. doi.org/10.1098/rsos.230447
- Navidi, P., **Saeedpour, S.**, Miandari, M., Ershadmanesh, S., Bahrami, B. (2023). Prosocial learning: Model-based or model-free? *PLoS One*. doi.org/10.1371/journal.pone.0287563

Poster Presentations

- **Inferring Ring-Attractor Dynamics in Head-Direction Neural Activity** 2025
PINTS (Paris Neuroscience, Theory and Systems) Conference, Paris.
- **Teaching by Description Can Transfer Rule Policies in a Deterministic Contextual Bandit** 2025
COSMOS (Computational Summer school on Modeling Social and collective behavior), Tokyo, Japan.

Honors, Awards and Scholarships

- **COSYNE Mentorship Travel Grant** 2026
Computational and Systems Neuroscience Conference.
- **Concours Normalien Étudiant** 2024
Highly selective French competitive exam granting Normalien student status at ENS-PSL, Cognitive Science.
- **COSMOS Summer School — Selected and Funded** 2025
Computational Summer school on Modeling Social and collective behavior, Tokyo, Japan.
- **Best Undergraduate Thesis Award** ([Link](#)) 2022
ECE School, University of Tehran.
- **ICPC World Finalist** ([Link](#)) Nov 2022
International Collegiate Programming Contest, Bangladesh. Coached team “aFunnyRandomPhrase”, University of Tehran.
- **Silver Medalist, 12th National Olympiad of Astronomy and Astrophysics (NOAA)** 2016

Teaching Experience

- **Neuromatch Academy — Computational Neuroscience** Remote
Teaching Assistant *Jul 2025 – Aug 2025*
- **University of Tehran** Tehran, Iran
Teaching Assistant *2019 – 2020*
Machine Learning (Fall 2020) · Engineering Probability & Statistics (Spring 2020, Fall 2019)
- **Allame Helli 1 High School** Tehran, Iran
Instructor, Astronomy & Astrophysics Olympiad *2017 – 2019*

Industry Experience

- **YUZ Games Studio – Cafe Bazaar** Tehran, Iran
Game Data Analyst *Apr. 2021 – Aug. 2022*
 - Employed statistical and machine learning approaches to analyze user behavior and optimize game levels to improve player experience.
- **People Analytics Team – Cafe Bazaar** Tehran, Iran
Human Resources Data Analyst *Jun. 2019 – Apr. 2021*
 - Utilized data-driven approaches to analyze employee profiles and optimize work conditions.

Service & Leadership

- **University of Tehran — Cognitive Neuroscience Competition for Students** Tehran, Iran
Executive Director, CNCS 2019 *Winter 2019*
 - Managed a team to execute the event and gathered speakers on different topics in cognitive science.
- **University of Tehran** Tehran, Iran
Performing Arts Club, Vice Chair *2019*
 - Managed activities of theater groups, shows, performances, workshops, and events around performing arts.